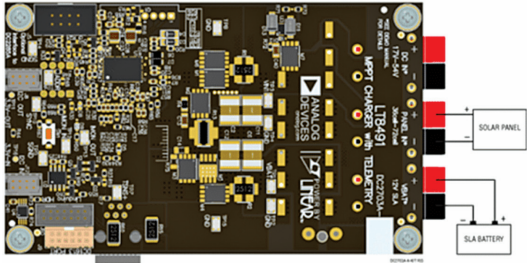


Interesting Reference Designs Of MPPT SOLAR CHARGER

Title of Reference Design	Battery Charger with MPPT	Solar MPPT Battery Charger
Image		
Key Attractions	The design provides efficient power management with MPPT functionality. It supports multiple battery chemistries and handles flexible input voltages. Real-time telemetry and programmable features make it ideal for solar and standalone charging.	The design optimises solar power extraction, supporting various battery types and solar panel configurations. It offers flexibility, scalability, and customisable features for efficient, safe charging in diverse applications and conditions.
Highlights	The Analog Devices reference design is a Li-ion battery charger with maximum power point tracking (MPPT) and telemetry, achieving 97% efficiency and supporting power sources, including solar panels up to 200W. Its buck-boost configuration allows input voltages from 17V to 54V, accommodating values below, above, or equal to the battery voltage, with a USB-to-I2C controller enabling communication. It is programmable and versatile and functions as a solar or standalone charger for multiple cell chemistries like Li-Ion, LiPo, and SLA. The LT8491 IC's on-chip logic ensures MPPT by tracking the maximum power point and periodically sweeping the input panel voltage to select the optimal power curve value. The charger includes a slave I2C interface for digital control, telemetry readouts, and status monitoring, compatible with the Windows-based GUI application 'simpleLT8491' for real-time control and monitoring of the charger and battery pack. The DC2703A-A-KIT can support higher current and voltage configurations by replacing input-side MOSFETs and capacitors with higher-rated components.	The reference design from Microchip gets the most energy from solar panels, even in changing conditions like light, shade, and temperature. It safely charges different types of batteries using predefined settings and works with various solar panels. This makes it useful for streetlights, road signs, portable generators, fans, and RV devices. The flexible design can handle power from less than 20W to over 400W. Engineers can customise it by changing capacitors, MOSFETs, and inductors. Extra features, such as monitoring battery power, checking temperature, and controlling fans, can be added or removed based on need. The charger works with solar panels that provide 15V to 60V and supports single or multiple connected panels. It uses MPPT to operate solar panels efficiently and has settings for different battery types. The system charges batteries step by step, adjusting current and voltage to keep them safe.
Applications	Applications include solar-powered battery chargers, multiple types of lead-acid battery charging, Li-ion battery chargers, and battery-equipped industrial or portable military equipment.	The reference design supports various panels, chemistries, and power ranges for outdoor applications.
OEM Brand	Analog Devices (ADI)	Microchip
URL	https://www.electronicsforu.com/news/new-products/reference-design-of-battery-charger-with-mppt	https://www.electronicsforu.com/electronics-projects/solar-mppt-battery-charger-reference-design